Holiday Practice Exam

VCE Algorithmics (HESS) · exam-style paper · 60 marks · suggested time 120 minutes

Sit it in one timed session. Answers at the back — mark yourself honestly.

Section A — Multiple choice (10 × 1 = 10 marks)

A1. $f(n) = 4n \log n + n^2$ is: A. $\Theta(n \log n)$ B. $\Theta(n^2)$ C. $\Theta(n^2 \log n)$ D. $O(n)$

A2. BFS on adjacency lists is $O(|V| + |E|)$ because: A. it visits each node $|E|$ times B. nodes queue once, edges scan at most twice C. the queue holds all edges D. it sorts the neighbours

A3. Which recurrence does the Master Theorem *not* apply to? A. $2T(n/2)+n$ B. $T(n/3)+1$ C. $T(n-1)+n$ D. $4T(n/2) + n^2$

A4. $T(n) = 4T(n/2) + n$ solves to: A. $\Theta(n)$ B. $\Theta(n \log n)$ C. $\Theta(n^2)$ D. $\Theta(n^2 \log n)$

A5. A problem is in NP iff: A. it cannot be solved in polynomial time B. a "yes" certificate can be verified in polynomial time C. it is NP-complete D. it is undecidable

A6. Hill climbing terminates when: A. the global optimum is reached B. temperature reaches zero C. no neighbour improves the current solution D. all solutions have been tried

A7. In simulated annealing, lowering T over time causes: A. more uphill moves accepted B. fewer uphill moves accepted C. larger neighbourhoods D. random restarts

A8. The Halting Problem is: A. NP-complete B. solvable for all programs by simulation C. undecidable D. decidable but intractable

A9. The Chinese Room argument concludes that: A. machines cannot pass the Turing Test B. running a program is insufficient for understanding C. weak AI is false D. the systems reply is correct

A10. Training error 2%, test error 35% indicates: A. underfitting B. overfitting C. a perfect model D. an unrepresentative training set only

Section B — Structured questions (50 marks)

B1. Complexity (10 marks)

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Algorithm Inventory(L):
    total ← 0
    Foreach x in L Do                 # Block 1
        total ← total + x
    Foreach i from 1 To n Do          # Block 2
        Foreach j from i+1 To n Do
            If L[i] = L[j] Do
                total ← total - 1
    size ← n                          # Block 3
    While size > 1 Do
        print(size)
        size ← size div 2
    Return total
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a.) Bound each block with justification. (6) b.) State the overall tightest bound. (1) c.) Using the formal definition, show Block 1's cost $f(n) = 3n + 2$ is $O(n)$. (3)

B2. Graph algorithms (8 marks)

A ride-share network has $|V|$ drivers and $|E|$ proximity links, adjacency lists.

a.) Justify BFS's $O(|V| + |E|)$ via the amortisation argument. (3)

b.) The company switches to an adjacency matrix. State the new bound for the same BFS and explain the change. (2)

c.) For one-source shortest paths on this sparse network, recommend Dijkstra-with-heap or Floyd–Warshall, with a quantitative reason. (3)

B3. Recursion (8 marks)

a.) Write the recurrence for MergeSort and solve it by the Master Theorem, naming the case. (3)

b.) Solve $T(n) = T(n-1) + n$, $T(1) = 1$, by expansion. (3)

c.) Explain why memoisation reduces naive Fibonacci from exponential to linear, referring to the call graph. (2)

B4. Intractability and heuristics (12 marks)

A logistics firm must route a single vehicle through 250 delivery points daily.

a.) Give the decision form of the problem and name its class membership (with the standard caveat). (2)

b.) Explain why brute force and simple pruning are both infeasible at this scale. (2)

c.) Design a heuristic approach: representation, construction, improvement, and escape from local optima. (4)

d.) The firm demands "the shortest possible route, guaranteed." Write the honest reply, including what can be offered instead. (4)

B5. Computability and AI (12 marks)

a.) Describe the structure of a Turing machine and state the Church–Turing thesis. (3)

b.) Prove the Halting Problem undecidable. (4)

c.) Distinguish strong from weak AI, and describe the Chinese Room argument's target and conclusion. (3)

d.) A forward-propagation question: inputs $x_1 = 1, x_2 = 1$; hidden h_1 (weights $0.4, 0.4$, bias -0.5), h_2 (weights $-0.6, 0.8$, bias -0.1); output (weights $1.0, 1.0$ from h_1, h_2, bias -1.5); step activations. Evaluate, showing working. (2)

Answers

Section A: 1 B · 2 B · 3 C · 4 C · 5 B · 6 C · 7 B · 8 C · 9 B · 10 B

B1 a) Block 1: $\Theta(n)$ — single pass. Block 2: $\Theta(n^2)$ — triangular double loop, $\frac{n(n-1)}{2}$ iterations. Block 3: $\Theta(\log n)$ — halving. b) $O(n^2)$. c) $c = 5, n_0 = 1$: $3n + 2 \leq 3n + 2n = 5n$ for $n \geq 1$.

B2 a) Each driver enqueued at most once ($|V|$); when dequeued, neighbours scanned — each link touched at most twice across the run ($2|E|$); total $O(|V| + |E|)$. b) $O(|V|^2)$ — finding neighbours in a matrix row costs $|V|$ regardless of degree. c) Dijkstra-heap: $O((|V| + |E|) \log |V|) \approx O(|V| \log |V|)$ sparse, vs FW $\Theta(|V|^3)$; also FW computes all pairs, which is more than required.

B3 a) $T(n) = 2T(n/2) + cn$; $n^{\log_2 2} = n = f$: case 2, $\Theta(n \log n)$. b) $T(n) = n + (n-1) + \cdots + 2 + T(1) = \frac{n(n+1)}{2} = \Theta(n^2)$. c) Only n distinct subproblems exist; memoisation computes each once ($O(1)$ combine), collapsing the exponential call tree to a chain.

B4 a) "Is there a route of length $\leq k$ through all 250 points?" — TSP-decision, NP-complete (hardness modulo P $\neq$ NP). b) $(249)! \approx 10^{490}$ tours; pruning cuts typical cases but the space remains astronomically large; exact solvers at this scale need techniques beyond the study design. c) e.g. permutation representation; nearest-neighbour construction; 2-opt improvement; simulated annealing (or restarts) for escape — any coherent quartet. d) No polynomial method guarantees optimality unless P = NP; offer: strong heuristic solutions with measured quality on small instances / lower-bound comparisons, plus the trade-off statement (guarantee surrendered for feasible runtime).

B5 a) Tape, head, finite states, transition table; thesis: every effectively computable function is Turing-computable — a definition-level claim, not provable. b) Standard proof: assume $H(P, x)$; build $D(P)$ inverting $H(P, P)$; $D(D)$ contradicts H either way; no H. c) Weak: simulates intelligence; strong: genuinely understands; Chinese Room targets strong AI — syntax manipulation without semantics, so running a program is insufficient for understanding. d) h_1: $0.4 + 0.4 - 0.5 = 0.3 > 0 \Rightarrow 1$; h_2: $-0.6 + 0.8 - 0.1 = 0.1 > 0 \Rightarrow 1$; out: $1 + 1 - 1.5 = 0.5 > 0 \Rightarrow \mathbf{1}$.